

DataMind: A Profile-Scoped Multi-Surface Data Plane for Agents

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Abstract

Tool-using agents increasingly operate over workspaces that combine documents, tables, graphs, and persistent state. These sources are commonly exposed as independent tools, leaving construction, revision visibility, provenance, concurrent maintenance, and failure recovery to application code. We present **DataMind**, a profile-scoped multi-surface data plane for agents. A *Build Agent* materializes raw workspace artifacts into validated document, database, graph, and memory surfaces and publishes a profile snapshot. A *Serving Agent* selects and composes surfaces for a request, returning evidence linked to sources and revisions. Shared surface contracts, receipts, snapshot publication, incremental updates, and bounded recovery connect the two paths. We evaluate the serving path with the auditable WorkSurface-Bench workload and the build path with WorkSurface-Build, then measure update visibility, concurrency, profile isolation, and source-fault recovery. The resulting build-serve-update-recovery lifecycle positions DataMind as a data management system for tool-using agents rather than as another vertical RAG application or collection of agent tools.

Keywords

data agents, data management, provenance, multi-surface retrieval, workspace, agent systems

1 Introduction

A tool-using language model rarely answers a serious enterprise question from its parameters alone. ReAct and Toolformer established the now-common pattern of interleaving model reasoning with calls to external tools [19, 24]; multi-agent runtimes such as AutoGen make those calls composable across an application [22]. In an enterprise workspace, the tools read a policy document, inspect a table, follow a file dependency, or recall a decision made earlier in a project. These artifacts form a workspace: a collection of heterogeneous data products that is private to a user or team and changes as work proceeds. The model is only one participant in this setting. A data system must decide how the workspace is represented, which representations are visible to a request, how an update is published, and how an answer can be traced back to its sources.

This setting is different from a conventional retrieval-augmented generation (RAG) application [15]. A document retriever can return passages, and graph-oriented RAG can organize evidence around entities or communities [7], but neither representation says whether

a database view was built from the same source revision. A text-to-SQL agent can execute a query, while systems such as DB-GPT and SiriusBI focus on natural language interaction with particular database or business-intelligence workloads [10, 23]; document-centric platforms such as RAGFlow focus on document parsing and retrieval [9]. They do not provide a policy for combining a table result with a document assertion or a dependency edge. A graph retriever can expose relationships, but it does not determine whether the graph is stale while the corresponding files are being edited. Existing agent runtimes and tool protocols make it easy to expose these components as endpoints [1, 22]; the semantics of the data exchanged between them remain application-specific.

The problem becomes visible when the agent must first decide where to look. Enterprise data-integration work has documented the gap between an LLM and the private, heterogeneous data products used by organizations [12], while pipeline-oriented systems such as AOP study how to compose operators over structured and unstructured sources [20]. WorkSurface-Bench isolates the corresponding serving problem: it evaluates enterprise tasks over document, table, graph, and cross-surface data, and reports routing, evidence, answer, and efficiency separately [16]. Its controlled settings show why surface selection is only one part of the problem: even when the required surface is known, the agent still has to perform the right operation and produce an answer supported by the returned evidence. A system that exposes all sources as unrelated tools cannot express which sources form one project, which revision should be read, or what evidence is safe to combine.

We study the following data-management problem:

Given a mutable workspace and a set of agent requests, how can a system materialize heterogeneous data surfaces, publish a consistent profile view, and serve that view to agents with bounded routing, provenance, update, and recovery semantics?

We present **DataMind**, a profile-scoped multi-surface data plane for tool-using agents. DataMind separates two roles that are often mixed together in application code. A *Build Agent* organizes raw workspace artifacts into data surfaces: parsed and indexed documents, relational projections, and dependency graphs. Profile and session memory is populated by the StoreAgent through the same surface contracts. A *Serving Agent* consumes a published profile view, selects the surfaces and operations relevant to a request, and returns values together with evidence receipts. Both roles use the same profile, surface contract, revision, provenance, and policy metadata.

The central abstraction is a profile snapshot. A profile is the isolation boundary for a user, project, or tenant. Its snapshot names the revisions of the surfaces that a request is allowed to observe. A surface manifest describes its schema, operations, source set, freshness, evidence type, access policy, and estimated cost. Build operations produce source-linked receipts; serving operations propagate those receipts into the answer trace. A build records a candidate descriptor

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and advances the logical publication pointer only after validation. The prototype blocks reads of surfaces touched by an active candidate and fails closed when a live-only provider can no longer satisfy a pinned request. This gives the system an explicit update boundary without requiring every physical backend to implement the same storage model. The design follows the provenance distinction between a result and its derivation [5, 8], but applies it to retrieval, SQL, graph traversal, and memory operations in one serving trace.

DataMind is not a new vector index, text-to-SQL model, graph algorithm, or language model. Its contribution is a data-plane boundary around the external state used during inference. The boundary allows an organization to retain its existing parser, database, vector store, graph store, or ETL job while exposing the resulting data products through a common profile and capability interface. It also gives local IDE agents and hosted runtimes the same data semantics; MCP, native loops, and SDKs are adapters at the edge of the plane.

We evaluate the two directions of the plane separately and together. WorkSurface-Bench is the main serving workload: it measures routing, evidence, answer quality, and efficiency under a fixed tool protocol [16]. WorkSurface-Build starts from raw workspace artifacts and measures materialization cost and the downstream value of the resulting surfaces. Additional workloads exercise incremental changes, concurrent requests, profile isolation, and failures of parsers, indexes, databases, and graphs. The goal is to measure not only whether an agent answers a question, but whether the answer remains reproducible as the workspace evolves.

This paper makes three contributions:

- We formulate workspace interaction as a profile-scoped multi-surface data-management problem with explicit contracts for surface operations, revision visibility, provenance, evidence, and policy.
- We design and implement DataMind, a build-serve runtime that materializes, publishes, routes, and updates document, relational, graph, and memory surfaces for tool-using agents, with explicit stale-read and candidate-build failure semantics.
- We combine an auditable multi-surface serving workload with build, update, concurrency, and failure experiments, separating agent quality from the data-system costs and correctness properties that support it.

Table 1 summarizes the qualitative positioning that motivates the comparison set. It is a capability matrix, rather than an accuracy result. DB-GPT and RAGFlow are retained as external baselines only on task subsets that their released systems support; quantitative comparisons use the same model, inputs, prompts, and output budget.

2 Problem Setting

2.1 Workspaces and data surfaces

A workspace W contains source artifacts $a_1, \dots, a_n$. An artifact can be a document, spreadsheet, database record, source file, generated report, or user-provided memory item. A *surface* is a materialized representation of a subset of W together with a family of operations that an agent can invoke. We consider four surfaces:

- a **document surface** with parsed blocks, chunks, embeddings, lexical indexes, and source locators;
- a **database surface** with typed tables or views and executable relational operations;
- a **graph surface** with entities, edges, and source annotations for dependency and relationship queries; and
- a **memory surface** with scoped records that persist project or user state across sessions.

The surfaces are complementary rather than mutually exclusive. A project report can yield passages for a document index, rows for a relational view, and an edge connecting the report to a code artifact. Treating each result as an independent tool loses this common origin. DataMind therefore associates source identifiers and checksums with derived records and exposes the relation between a result and the surface that produced it.

2.2 Profiles and snapshots

A profile p binds a workspace, source roots, access policy, surface manifests, and persistent state. Profiles provide the boundary required for local projects and multi-tenant services: a request in profile p must not implicitly read a source or memory item belonging to p' . A profile may also contain session state, but session state is a narrower scope than profile state and does not change the profile's durable data products.

For each surface s , a build produces a revision r_s . A snapshot σ_p is a persistent publication record containing a vector of visible revisions:

$$\sigma_p = \langle r_{\text{doc}}, r_{\text{db}}, r_{\text{graph}}, r_{\text{memory}} \rangle.$$

A serving request is evaluated against (p, σ_p) . The publication record is advanced only after candidate validation. The built-in prototype providers currently expose a live view rather than historical artifact files. Reads for surfaces touched by an active candidate are blocked, and if a request's pinned record is superseded while it is running, the runtime fails closed instead of silently combining revisions. This explicit boundary keeps the logical snapshot contract useful while making the provider limitation observable and measurable.

2.3 Surface contracts

Each surface exposes a manifest

$$M_s = (\text{schema}, \text{ops}, \text{sources}, \text{revision}, \text{freshness}, \text{evidence}, \text{policy}, \text{cost}).$$

The schema describes the shape of values and locators. Operations describe what the Serving Agent can do, such as lexical or vector retrieval, SQL execution, or graph traversal. Freshness and revision describe when the result was produced. Evidence describes how a result can be cited: a document span, a query and returned rows, or a graph path with source edges. Cost summarizes latency, token transfer, and optional external-service cost. Policy states the profile and operation constraints.

The manifest is used in both directions. The Build Agent uses it to declare what a materialization will contain and which source artifacts it covers. The Serving Agent uses it to eliminate operations whose schema, scope, or evidence type cannot satisfy a request. This is a data-system interface: it keeps the model-facing operation

Table 1: Qualitative positioning against representative data and agent systems. A check mark denotes first-class support, ~ partial or product-specific support, and a cross a missing support or a non-goal.

System	Profile	Docs	SQL	Graph	Memory	Wiki	Update	Pipeline	IDE	Guards
DB-GPT [6, 23]	~	✓	✓	✓	~	✗	~	~	~	~
RAGFlow [9]	~	✓	~	✓	✓	✗	✓	~	~	~
Dify [14]	~	✓	~	✗	~	✗	✓	~	~	~
AnythingLLM [17]	~	✓	✓	✗	✓	✗	~	~	✗	✗
LangChain [4]	✗	~	~	~	~	✗	✗	✗	✗	~
LangGraph [13]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓
DataMind	✓	✓	✓	✓	✓	✓	✓	~	✓	✓

stable while allowing a surface to change its physical index or storage implementation.

2.4 Receipts and evidence

A write receipt records the profile, source reference, surface operation, revision, item count, and status. A read evidence receipt records the profile, snapshot, surface, source identifier, locator, and returned value. A response may contain several receipts when a question requires a document, a database, and a graph operation. The receipt set is the system’s explanation of which materialized facts were available to the generator; it is not a post-hoc similarity score.

This design follows the database provenance distinction between a result and its derivation [5, 8], while adapting the unit of provenance to heterogeneous Agent operations. The prototype does not claim tuple-level provenance for every physical backend. It does claim that each DataMind operation identifies the source and revision it exposes, and that the request trace preserves the operations used to form the answer.

2.5 Build, serve, and maintenance goals

The system is designed around five goals:

- (1) **Materialization**: organize heterogeneous artifacts into complementary surfaces while retaining source identity;
- (2) **Serving**: choose and compose surface operations without requiring the model to copy large intermediate results through its context;
- (3) **Visibility**: make updates observable at a defined revision and keep a request on its selected snapshot;
- (4) **Isolation**: carry profile and policy through every read and write; and
- (5) **Recovery**: distinguish an empty result from an unavailable surface and resume or fall back without silently losing evidence.

3 DataMind Data Plane

DataMind is the complete data plane between a mutable workspace and a tool-using Agent. It has two logically separate paths that share the same profile, surface contracts, provenance, policy, and revision semantics. Figure 1 is the serving view: it shows how a request is routed over already materialized surfaces and returned with evidence. The corresponding ingestion path is DataMind-Build,

shown in Figure 2; it is the Build Agent and its publication pipeline, not a second product or a different data model.

3.1 Workspace-to-surface materialization

DataMind-Build makes a workspace usable. Its Build Agent begins with an inspection of profile-allowed roots and an artifact inventory. It then selects a parser and produces provider-neutral blocks with source locators. PDF and office documents may use a hosted document parser; a local PDF parser is retained as a deterministic fallback. Tables are normalized into rows and columns, while source and dependency files can contribute graph entities and edges. The parser choice is recorded because changing a parser can change a surface revision even when the source checksum is unchanged.

The next stage projects the extracted blocks into one or more physical surfaces. Document blocks are chunked and indexed for hybrid retrieval. Tables are written to a relational backend with a schema that can be inspected before query execution. Graph records are written with source annotations so a path can be traced to the files that induced its edges. The current build path materializes document, database, and graph surfaces; memory and skills remain profile-scoped runtime surfaces populated by their own write tools. The Build Agent is allowed to produce only the surfaces selected by the profile configuration; it does not assume that every workspace needs every representation.

Each projection emits a receipt and is validated before publication. Validation checks include source coverage, schema shape, duplicate handling, and surface-specific invariants such as parseable SQL tables or resolvable graph locators. A successful build produces a manifest and a candidate snapshot. The candidate record is outside the normal publication path until validation and the single pointer update. During assembly, the read guard blocks the touched surfaces so live-only providers cannot leak a partial candidate. The current prototype does not claim that every provider stages physical files transactionally; it reports an explicit failure when a live-only provider can no longer satisfy a pinned record.

3.2 Surface-to-agent serving

The Serving Agent receives a request together with its profile and snapshot. It first inspects the available surface manifests. A request about a policy clause can be answered from the document surface; a request requiring aggregation should use a database operation; a dependency question needs the graph. A cross-surface request

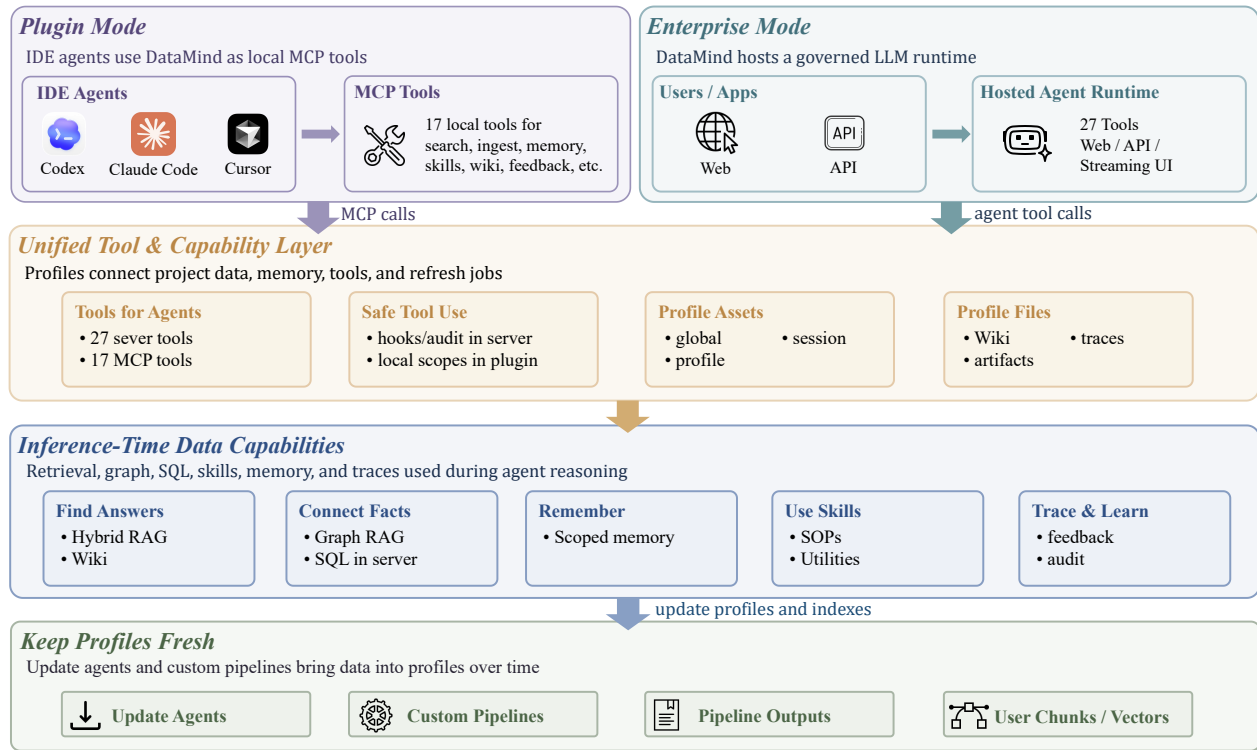


Figure 1: DataMind serving path. The Serving Agent routes a request over profile-scoped document, database, graph, memory, and skill surfaces and returns an answer with evidence. The profile, policy, revision, and receipt interfaces are shared with DataMind-Build.

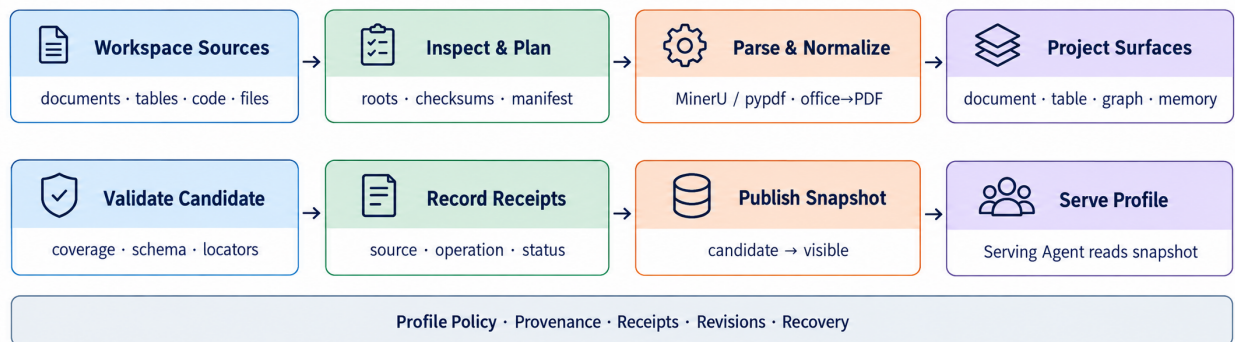


Figure 2: DataMind-Build ingestion path. The Build Agent inventories a profile’s workspace, plans surface candidates, parses and normalizes sources, projects records, validates candidate revisions, and publishes a profile snapshot. The candidate descriptor remains unpublished until validation; reads of touched surfaces are blocked during candidate assembly, and providers that do not support historical artifacts fail closed if a pinned request becomes stale.

produces a plan that includes the minimum set of operations whose evidence types can jointly support the answer.

The planner does not treat all tools as equivalent. It uses operation schemas to decide which arguments are legal and source and revision metadata to avoid mixing incompatible views. Manifests

also expose cost and freshness for analysis and future planning; the current prototype does not implement a cost-based optimizer. The model remains responsible for interpreting the request and synthesizing natural language, but the data plane controls the values and evidence that can enter that synthesis. Intermediate results can be referenced by receipt rather than copied verbatim through the model context. This distinction is important for tables and graph neighborhoods that are large relative to a prompt.

When an operation fails, the Serving Agent preserves the distinction between an empty result and an unavailable capability. An empty retrieval result is evidence that the surface returned no matching item; an unavailable graph store is an execution failure. The model may issue another tool call, but the runtime does not claim an automatic cross-surface recovery planner. The trace records the failed operation and the receipts used by any final answer.

3.3 Updates and version visibility

A source update is represented as a new source fingerprint and a new profile revision. The ingestion ledger makes a repeated write idempotent: the same normalized tool call and source checksum returns an unchanged receipt rather than adding a second copy. A changed source produces a new receipt and updates the surface-specific materialization. This ledger is deliberately separate from the physical index so that a profile can be audited even when a backend is replaced.

For a multi-surface build, the system records a candidate and publishes a snapshot after validation. Serving requests pin the publication record at request start. A provider that can open that record may continue on it; the built-in live-only providers instead fail closed when publication advances. While a candidate is being assembled, reads of surfaces touched by that candidate are blocked until validation and publication complete. This is the visibility rule used in the update experiments: stale reads are counted as explicit failures, never as apparently successful mixed views. Physical staging and historical reads are provider-level extensions beyond the current prototype and are not assumed by the core claim.

3.4 Access control and profile isolation

DataMind separates update visibility from access control. Updates determine which revision a request observes; access control determines which sources and operations that request may access. Policy is enforced at dispatch, where a proposed model action becomes an operation on data. Path policies resolve symlinks and restrict file and CSV access to profile roots and configured extra roots. SQL policies distinguish read-only queries from destructive statements and can deny, request confirmation, or rewrite a call. Audit records include allowed, denied, and confirmation-required operations. The profile identifier is carried through these checks, the capability handler, the receipt, and the final trace.

This boundary complements rather than replaces database authorization. A backend still enforces its own permissions; DataMind prevents an Agent from reaching an operation that the profile policy has not exposed. The same mechanism also applies to a local MCP client and to the hosted runtime, which keeps safety behavior independent of the model SDK.

4 Prototype

DataMind is implemented as a protocol-oriented Python runtime. A registry maps a surface operation to a handler, input schema, access mode, and metadata. The agent loop sees the registry rather than a backend-specific object. The current prototype contains a hybrid document retriever, a SQLite relational backend, a NetworkX-backed graph store, and a SQLite-backed memory provider. Each provider can be replaced while preserving the profile and receipt interfaces.

The implementation has two logical agents. The StoreAgent owns writes and returns structured ingestion receipts. It routes file, folder, text, CSV, triple, skill, and memory operations through the ingestion ledger. The RetrieveAgent is read-only during normal chat and returns an answer, evidence, surfaces used, and a tool trace. This division makes the read/write boundary explicit and gives the evaluation a direct way to compare a mutable pipeline with a snapshot-based one.

Document format adapters cover plain text, Markdown, HTML, JSON, XML, source code, and PDF. Legacy Office documents (DOC and PPT) are converted to PDF via LibreOffice; DOCX and PPTX use native extractors in the current prototype. Tabular files (CSV/TSV and XLS/XLSX) are loaded into typed relations. The original file remains the lineage source. MinerU is accessed through its hosted API when configured, and pypdf provides a local PDF fallback. These choices affect build quality and cost, but are not themselves the data-plane contribution.

The runtime can be used through a native model loop, a provider SDK loop, or an MCP adapter. The adapter used by an IDE exposes the same profile directory and surface operations as the hosted service. This makes the local plugin a deployment surface for the data plane rather than a separate system. The implementation section intentionally does not claim that every backend or external pipeline is production-hardened; those claims are tested only when the corresponding end-to-end path is included in the experiments.

5 Evaluation

The evaluation is organized around the lifecycle in Figure 1: materialize workspace data, serve requests over a published view, maintain the view under changes, and recover when a component fails. We use four research questions:

- **RQ1:** Does surface-aware serving improve route, evidence, answer quality, and efficiency?
- **RQ2:** What is the cost of building multiple surfaces, and when is that cost amortized by later queries?
- **RQ3:** What latency, throughput, and memory overhead does DataMind add under concurrent serving?
- **RQ4:** Can DataMind recover from surface and build failures without returning untraceable answers?

5.1 Workloads and baselines

The main serving workload is WorkSurface-Bench. It contains auditable tasks whose required evidence is grounded in document spans, executable table queries, and annotated graph sources [16]. We adapt the canonical document, table, and graph representations to DataMind surface contracts and verify the adapter before any

Table 2: RQ1 main serving results on WorkSurface-Bench.

Model	Condition	Route-F1↑	Evidence↑	Answer↑	Efficiency↑	Tokens↓	p95 ms↓
GPT-4o-mini	ReAct-all	TBD	TBD	TBD	TBD	TBD	TBD
GPT-4o-mini	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD
DeepSeek-V4-Pro	ReAct-all	TBD	TBD	TBD	TBD	TBD	TBD
DeepSeek-V4-Pro	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD
Gemini-3.1-Pro	ReAct-all	TBD	TBD	TBD	TBD	TBD	TBD
Gemini-3.1-Pro	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD
GPT-5.5	ReAct-all	TBD	TBD	TBD	TBD	TBD	TBD
GPT-5.5	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD

Table 3: RQ1 answer quality by WorkSurface-Bench task type.

Condition	RAG	Table	Graph	Cross-surface	Evidence
ReAct-all	TBD	TBD	TBD	TBD	TBD
Naive-router	TBD	TBD	TBD	TBD	TBD
DataMind-full	TBD	TBD	TBD	TBD	TBD
Gold-hint/all	TBD	TBD	TBD	TBD	TBD

model run. The normal serving agent receives no gold surface label. Gold-constrained settings are reported only as upper bounds.

We compare DataMind with the benchmark’s no-tool, document-only, single-surface, and all-tools ReAct settings. We also implement a simple manifest router that receives the same surface descriptions but does not use DataMind’s snapshot, evidence, or receipt path. All comparisons use the same model backbone, prompt budget, tool schemas, task order, and temperature. The backbone sweep follows WorkSurface-Bench (GPT-4o-mini, DeepSeek-V4-Pro, Gemini-3.1-Pro, and GPT-5.5); the DataMind build and dynamic-system studies fix GPT-5.5 to isolate data-plane effects. We report repeated-run means and confidence intervals rather than a single trajectory.

We additionally retain the two external systems from the earlier report as matched-subset baselines. DB-GPT is evaluated on database and structured-query tasks, and RAGFlow on document/RAG tasks, using the same model, source files, questions, prompt budget, and decoding settings as DataMind. They are not assigned a score on graph or cross-surface tasks that their deployment does not support. If a released version cannot be run under these controls, we report the capability comparison in Table 1 and omit a numerical claim rather than conflate system coverage with answer quality.

5.2 RQ1: Serving quality and efficiency

The primary metrics are Route F1, Evidence, Answer, and Efficiency. We also measure the number of accessed surfaces, operation calls, invalid calls, irrelevant surface calls, input/output tokens, and end-to-end latency. The analysis separates single-surface and cross-surface tasks. This distinction prevents a system from appearing efficient merely because it avoids the hard cases that require evidence from more than one surface.

RQ1 reports serving quality, routing behavior, evidence coverage, and runtime cost under a fixed single-version workspace. Snapshot publication and parser fallback are evaluated in the dynamic update and fault-injection experiments, respectively, because a static benchmark with one valid revision and no injected failures cannot identify their effects. If the full system improves answer quality but increases unnecessary exploration, the additional tool and token measurements make that trade-off explicit.

Table 4: RQ1 matched-subset comparison with external systems.

System	Subset	Route-F1	Evidence	Answer	p95 ms	Errors
DB-GPT	Table/SQL	TBD	TBD	TBD	TBD	TBD
RAGFlow	RAG/document	TBD	TBD	TBD	TBD	TBD
DataMind-full	Same subset	TBD	TBD	TBD	TBD	TBD

Table 5: RQ2 build cost and materialization coverage.

Strategy	Time s↓	API calls↓	Storage MB↓	Coverage↑	Provenance↑	Validation failures↓
Document-only	TBD	TBD	TBD	TBD	TBD	TBD
Table-only	TBD	TBD	TBD	TBD	TBD	TBD
Graph-only	TBD	TBD	TBD	TBD	TBD	TBD
Eager-all	TBD	TBD	TBD	TBD	TBD	TBD
Heuristic-demand	TBD	TBD	TBD	TBD	TBD	TBD
DataMind-build	TBD	TBD	TBD	TBD	TBD	TBD

Table 6: RQ2 downstream serving value after materialization.

Strategy	<i>N</i>	Answer	Evidence	Efficiency	p95 ms	Cost/query
Document-only	100	TBD	TBD	TBD	TBD	TBD
Eager-all	100	TBD	TBD	TBD	TBD	TBD
DataMind-build	100	TBD	TBD	TBD	TBD	TBD

5.3 RQ2: Build cost and amortization

WorkSurface-Build begins with raw workspace artifacts and produces the surfaces used by a fixed Worker. We compare document-only, database-only, graph-only, an unmanaged all-surface build, heuristic demand selection, and DataMind’s managed build. The unmanaged all-surface condition uses the same configured document, database, and graph surfaces, parsers, and physical backends as DataMind but omits manifest-driven validation, receipts, snapshot publication, and revision management. Build metrics include wall clock time, parser and embedding calls, storage, source coverage, provenance completeness, duplicate handling, and failed-stage recovery. After each build, the same Worker answers the serving workload so that materialization quality is not conflated with Agent behavior.

For N subsequent queries, we report

$$C(N) = C_{\text{build}} + N \cdot C_{\text{serve}}.$$

The break-even point is the smallest N for which the extra build cost of a multi-surface representation is recovered by its serving quality or cost benefit. We also report a quality-cost curve instead of only a single aggregate score. This is the experiment that connects WorkSurface-Build to the DataMind claim: materialization is useful only when its cost produces durable serving value.

5.4 RQ3: Runtime overhead and concurrency

RQ3 measures the system cost of DataMind’s serving path. We run the same representative workload with the same model, prompt, tool schemas, and surface contents under two conditions. *DataMind no-hooks* disables the PathAllowlist, destructive-SQL, and audit hooks while retaining the serving runtime. *DataMind full* uses the default configuration, including hooks, evidence, snapshot metadata, and audit logging. If an external DB-GPT implementation can be reproduced with the same model and data, we report it as

Table 7: Pilot ingestion component sensitivity from the earlier report on its MS MARCO subset. These values must be rerun before being used as a primary WorkSurface-Build result.

System	Chunker	Embedder	NDCG@10↑	Recall@10↑	MRR↑
DataMind auto	built-in	Qwen v2	0.9625	0.9920	0.9533
DataMind BYOP	sentence	Qwen v2	0.9641	0.9960	0.9543
DataMind BYOP	sentence	Qwen v3	0.9859	0.9960	0.9828
DataMind BYOP	sentence	Qwen v4	0.9861	1.0000	0.9814
DB-GPT	built-in	Qwen v4	0.9834	0.9980	0.9785

Table 8: RQ3 end-to-end latency, throughput, and memory under concurrency. Concurrency-one values are the pilot measurements from the earlier report; the remaining rows are reserved for the controlled rerun.

System	Concurrency	N	Errors↓	p50 ms↓	p95 ms↓	p99 ms↓	QPS↑	Peak RSS MB↓
DataMind no-hooks	1	60	2	9373.8	18329.4	24302.7	0.0917	193.5
DataMind full	1	60	2	8458.0	15413.4	19560.0	0.1012	188.1
DataMind no-hooks	20	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DataMind full	20	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DataMind no-hooks	100	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DataMind full	100	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DB-GPT (optional)	1	60	0	7164.1	11660.2	13839.5	0.1297	194.0

an additional reference; otherwise we do not attribute differences to an unmatched external system.

We run 60 or 200 tasks at concurrency 1, 5, 20, 50, and 100, with at least three repetitions per condition. We report errors, p50/p95/p99 end-to-end latency, QPS, peak RSS, tool calls, input/output tokens, and a model-versus-tool latency breakdown. QPS is completed tasks divided by wall-clock time. Error rate is reported separately from latency averages. The comparison quantifies the runtime cost of DataMind’s policy, audit, evidence, and version metadata while holding model behavior constant.

5.5 RQ4: Failure recovery

We inject parser failures, MinerU timeouts, embedding-service failures, empty document results, database and SQL failures (timeouts, connection failures, locked or corrupted views), graph unavailability, interrupted builds, and corrupted derived artifacts. For each fault, we record whether the request completes, whether its answer is correct, whether the evidence remains auditable, and how long the structured failure or parser fallback takes. We compare retry-only, fixed-path structured errors, and DataMind’s structured errors plus the MinerU-to-pypdf parser fallback and candidate/read guards. DataMind does not automatically substitute one failed surface for another.

The fault matrix distinguishes unavailable data from empty data. The parser fallback is counted as successful only if the resulting evidence satisfies the task requirement; a plausible answer without a valid receipt is a failure. Surface outages and stale-provider blocks are counted as explicit failures, not as successful fallbacks. This prevents the recovery experiment from rewarding a model for guessing after a source outage.

Table 9: RQ4 recovery after injected surface and build faults.

Fault	System	Completion	Answer	Auditable	Structured error	Fallback/block	Recovery ms
Parser/MinerU failure	Retry-only	TBD	TBD	TBD	TBD	TBD	TBD
Parser/MinerU failure	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD
Database/SQL timeout	Retry-only	TBD	TBD	TBD	TBD	TBD	TBD
Database/SQL timeout	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD
Active candidate read	Retry-only	TBD	TBD	TBD	TBD	TBD	TBD
Active candidate read	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD
Graph unavailable	Retry-only	TBD	TBD	TBD	TBD	TBD	TBD
Graph unavailable	DataMind-full	TBD	TBD	TBD	TBD	TBD	TBD

Table 10: Pilot recovery results from the earlier report. The denominator is the affected recoverable slice for each injected fault.

System	KB empty	Graph down	DB timeout	Partial corruption	Overall
Fixed pipeline	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)
DataMind live	60.0% (3)	60.7% (17)	70.6% (12)	62.5% (25)	63.3% (57)

Table 11: Pilot profile-scoped memory isolation diagnostic from the earlier report.

Setting	Memory leakage↓	Memory recall↑	Answer leakage↓	Answer recall↑	Mixed answer↓
Flat namespace	30%	70%	68%	31%	63%
DataMind three-scope	0%	100%	0%	100%	0%

Table 12: Pilot tool-dispatch SQL policy diagnostic from the earlier report. Ask-user decisions count as blocked until confirmation.

System	Unsafe recall↑	FP rate↓	Recall-FP↑
Direct execution	0%	0%	0%
Keyword guard	8%	0%	8%
DB-GPT	82%	56%	26%
DataMind hooks	100%	0%	100%

The following two diagnostics preserve the earlier report’s scope-isolation and tool-policy experiments. They are reported as correctness checks rather than as additional claims about the serving quality benchmark.

5.6 Ablation and overhead

A final experiment isolates runtime overhead on a local single-surface workload. We measure manifest lookup, receipt creation, policy checks, snapshot selection, and audit logging separately from model latency. Cold-start and warm-cache runs are randomized, repeated, and reported with p50/p95/p99 latency and throughput. This experiment is not intended to show that DataMind is faster than a fixed RAG pipeline; it quantifies the cost of the semantics required for profiles, evidence, and recovery.

6 Related Work

6.1 RAG, graph retrieval, and memory

RAG grounds generation in an external corpus [15], while GraphRAG adds graph structure for query-focused retrieval and summarization [7]. Modular retrieval toolkits such as FlashRAG make it easier to compare retrievers and indexing choices [11]. These works improve how one representation is searched. DataMind addresses the boundary between several representations:

it records how a source contributes to a document, relational, graph, or memory surface and exposes the resulting revision and evidence metadata to the Agent.

Long-term-memory systems such as MemGPT treat durable state as a first-class part of an Agent architecture [18]. DataMind’s memory surface is narrower: it provides profile and session scopes and integrates memory writes with the same receipt and policy path as other surfaces. It does not propose a new memory retrieval algorithm.

6.2 LLM systems for data and data lakes

DB-GPT demonstrates an end-to-end system for natural-language interaction with data [23], while SiriusBI focuses on industrial business intelligence, multi-round dialogue, and domain-adaptive SQL generation [10]. Pneuma studies LLM-based representation and retrieval for tabular data discovery [2]. NeurDB explores an AI-powered autonomous database architecture [25]. These systems target particular data-management experiences or database workloads. DataMind is complementary: its unit of management is a profile containing multiple data surfaces, and its evaluation includes the construction and maintenance of those surfaces rather than only a database interaction task.

AOP is closer in spirit because it composes semantic operators over structured, semi-structured, and unstructured data and studies pipeline orchestration and optimization [20]. DataMind differs in the lifecycle boundary it makes explicit. AOP primarily optimizes an online semantic pipeline; DataMind tracks the materialized surfaces, profile snapshot, source revision, and receipts that connect an offline build to later serving and updates.

Enterprise data integration work has also shown that public benchmarks can overestimate LLM performance on private organizational data and has proposed annotations and ontology synthesis to close that gap [12]. WorkSurface-Bench follows this enterprise-workload motivation, while DataMind uses the workload to evaluate a data-plane runtime rather than a data-integration model.

6.3 Agent–database interfaces and workflow reliability

BridgeScope decomposes database access into fine-grained tools, aligns tools with database privileges, and provides a proxy for transferring intermediate results without routing them through the LLM [21]. These are important database-side mechanisms. DataMind generalizes the boundary to several surfaces and adds profile snapshots, build-time materialization, and source-linked evidence across those surfaces.

SagaLLM applies context management, validation, and compensating transactions to multi-agent planning workflows [3]. DataMind shares the concern for durable state and recovery but has a different state boundary: it manages the data products that an Agent reads and writes, rather than the business actions of a multi-agent plan. We therefore make a narrower claim than full ACID execution for arbitrary Agent actions.

Classical provenance and workflow systems provide the conceptual basis for source lineage, derivation, and replay [5, 8]. DataMind’s challenge is to carry those ideas across vector retrieval, SQL,

graph traversal, and memory operations while exposing enough metadata for a language model to select an operation.

7 Conclusion

DataMind treats a workspace used by a tool-using Agent as a managed collection of data surfaces. The Build Agent organizes raw artifacts into document, relational, graph, and memory representations. The Serving Agent selects and combines those representations for a request. Profiles, snapshots, surface contracts, receipts, policy checks, and recovery connect the two directions.

This build–serve–update–recovery view gives a concrete data-management interpretation to multi-surface Agent workloads. WorkSurface-Bench measures whether the serving path obtains the right surfaces and evidence; WorkSurface-Build measures the cost and value of constructing them; dynamic experiments measure whether the resulting view remains correct as the workspace changes. The central claim of the paper is therefore about the data plane beneath the Agent, not about a particular model, parser, SDK, or tool list.

References

- [1] Anthropic and Model Context Protocol Contributors. 2025. Model Context Protocol Specification: Server Tools (Revision 2025-06-18). <https://modelcontextprotocol.io/specification/2025-06-18/server/tools>. Accessed 2026-05-19.
- [2] Muhammad Imam Luthfi Balaka, David Alexander, Qiming Wang, Yue Gong, Adila Krisnadhi, and Raul Castro Fernandez. 2025. Pneuma: Leveraging LLMs for Tabular Data Representation and Retrieval in an End-to-End System. *Proceedings of the ACM on Management of Data* 3, 3 (2025). doi:10.1145/3725337
- [3] Edward Y. Chang and Longling Geng. 2025. SagaLLM: Context Management, Validation, and Transaction Guarantees for Multi-Agent LLM Planning. *Proceedings of the VLDB Endowment* 18, 12 (2025), 4874–4886. doi:10.14778/3750601.3750611
- [4] Harrison Chase and LangChain Contributors. 2026. LangChain: Retrieval Documentation. <https://docs.langchain.com/oss/python/langchain/retrieval>. Accessed 2026-05-28.
- [5] James Cheney, Laura Chiticariu, and Wang-Chiew Tan. 2009. Provenance in Databases: Why, How, and Where. *Foundations and Trends in Databases* 1, 4 (2009), 379–474. doi:10.1561/19000000006
- [6] DB-GPT Contributors. 2026. DB-GPT Documentation. <https://docs.dbgpt.cn/docs/overview/>. Accessed 2026-05-28.
- [7] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. <https://arxiv.org/abs/2404.16130>. arXiv:2404.16130 [cs.CL]
- [8] Todd J. Green, Grigoris Karvounarakis, and Val Tannen. 2007. Provenance Semirings. In *Proceedings of the 26th ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*. 31–40. doi:10.1145/1265530.1265535
- [9] InfiniFlow. 2026. RAGFlow: Open-source RAG Engine Based on Deep Document Understanding. <https://github.com/infiniflow/ragflow>. Accessed 2026-05-28.
- [10] Jie Jiang, Haining Xie, Siqu Shen, Yu Shen, Zihan Zhang, Meng Lei, Yifeng Zheng, Yang Li, Chunyou Li, Danqing Huang, Yinjun Wu, Wentao Zhang, Bin Cui, and Peng Chen. 2025. SiriusBI: A Comprehensive LLM-Powered Solution for Data Analytics in Business Intelligence. *Proceedings of the VLDB Endowment* 18, 12 (2025), 4860–4873. doi:10.14778/3750601.3750610
- [11] Jiajie Jin, Yutao Zhu, Zhicheng Dou, Guanting Dong, Xinyu Yang, Chenghao Zhang, Tong Zhao, Zhao Yang, and Ji-Rong Wen. 2025. FlashRAG: A Modular Toolkit for Efficient Retrieval-Augmented Generation Research. In *Companion Proceedings of the ACM Web Conference 2025 (WWW '25 Companion)*. Association for Computing Machinery, 737–740. arXiv:2405.13576 [cs.CL] doi:10.1145/3701716.3715313
- [12] Moe Kayali, Fabian Wenz, Nesime Tatbul, and Catagay Demiralp. 2025. Mind the Data Gap: Bridging LLMs to Enterprise Data Integration. In *Conference on Innovative Data Systems Research (CIDR)*. <https://www.vldb.org/cidrdb/2025/mind-the-data-gap-bridging-llms-to-enterprise-data-integration.html>
- [13] LangChain AI. 2024. LangGraph: Building Stateful, Multi-ACTOR Applications with LLMs. <https://github.com/langchain-ai/langgraph>. GitHub repository; accessed 2026-05-28.
- [14] LangGenius. 2026. Dify: Production-ready Platform for Agentic Workflow Development. <https://github.com/langgenius/dify>. Accessed 2026-05-28.

- [15] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33. 9459–9474. arXiv:2005.11401
- [16] Hao Liang, Meiyi Qiang, Sizhe Qiu, Linzhuang Sun, and Wentao Zhang. 2026. WorkSurface-Bench: Benchmarking Enterprise Agents on Multi-Surface Knowledge Routing. <https://arxiv.org/abs/2607.25765>. arXiv:2607.25765 [cs.DB]
- [17] Mintplex Labs. 2026. AnythingLLM: All-in-one AI Application for Local LLMs, RAG, and Agents. <https://github.com/Mintplex-Labs/anything-llm>. Accessed 2026-05-28.
- [18] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. <https://arxiv.org/abs/2310.08560>. arXiv:2310.08560 [cs.AI]
- [19] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. 68539–68551. arXiv:2302.04761
- [20] Jiayi Wang and Guoliang Li. 2025. AOP: Automated and Interactive LLM Pipeline Orchestration for Answering Complex Queries. In *Conference on Innovative Data Systems Research (CIDR)*. <https://www.vldb.org/cidrdb/2025/aop-automated-and-interactive-llm-pipeline-orchestration-for-answering-complex-queries.html>
- [21] Lianggui Weng, Dandan Liu, Rong Zhu, Bolin Ding, and Jingren Zhou. 2026. BridgeScope: A Universal Toolkit for Bridging Large Language Models and Databases. In *Conference on Innovative Data Systems Research (CIDR)*. <https://www.vldb.org/cidrdb/2026/bridgescope-a-universal-toolkit-for-bridging-large-language-models-and-databases.html>
- [22] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryan W. White, Doug Burger, and Chi Wang. 2024. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. In *Conference on Language Modeling (COLM)*. arXiv:2308.08155
- [23] Siqiao Xue, Danrui Qi, Caigao Jiang, Fangyin Cheng, Keting Chen, Zhiping Zhang, Hongyang Zhang, Ganglin Wei, Wang Zhao, Fan Zhou, Hong Yi, Shaodong Liu, Hongjun Yang, and Faqiang Chen. 2024. Demonstration of DB-GPT: Next Generation Data Interaction System Empowered by Large Language Models. *Proceedings of the VLDB Endowment* 17, 12 (2024), 4365–4368. arXiv:2404.10209 doi:10.14778/3685800.3685876
- [24] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations (ICLR)*. arXiv:2210.03629
- [25] Zhanhao Zhao, Shaofeng Cai, Haotian Gao, Hexiang Pan, Siqi Xiang, Naili Xing, Gang Chen, Beng Chin Ooi, Yanyan Shen, Yuncheng Wu, and Meihui Zhang. 2025. NeurDB: On the Design and Implementation of an AI-powered Autonomous Database. In *Conference on Innovative Data Systems Research (CIDR)*. <https://www.vldb.org/cidrdb/2025/>